36-662 Statistical Machine Learning
Final Report

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1 Introduction

This project uses flight data from the U.S. Bureau of Transportation Statistics of the U.S. Department of Transportation, with a focus on flights to and from Pittsburgh International Airport (PIT) during 2023–2025. The dataset contains information on scheduled flights, as well as their actual departure time, delay status and causes of delay. It also includes several variables indicating the details of the flight, such as the carrier, tail number, destination, and whether the flight experienced diversions or cancellations.

The primary objective is to predict whether a departing flight from Pittsburgh will experience a departure delay (defined as a delay of 15 minutes or more). Predictions must be made using only information available prior to the scheduled departure time. While the dataset includes both arriving and departing flights, the analysis focuses on departing flights, with arriving flights used to construct additional predictive features.

This is a supervised classification problem, requiring careful handling of time-dependent information to avoid using data that would not be available at the time of prediction. Such predictions are valuable for airlines and airport operations, as they enable better planning, resource allocation and communication with passengers in the presence of potential delays.

2 Exploration

Prior to any analysis, we explored the delay rates of flights departing from Pittsburgh. From our prior knowledge about flights, we expected that certain features already contained in the dataset would have some relationship with whether a flight was delayed, such as day of the week and time of day. As expected, Figure 1 shows that flights on Fridays, weekends and Mondays are delayed at higher rates, likely due to a higher volume of passengers traveling on those days. Similarly, Figure 2 shows that flights later in the day are delayed more often than flights earlier in the day, usually because planes make multiple trips per day and one delay can affect all remaining trips.

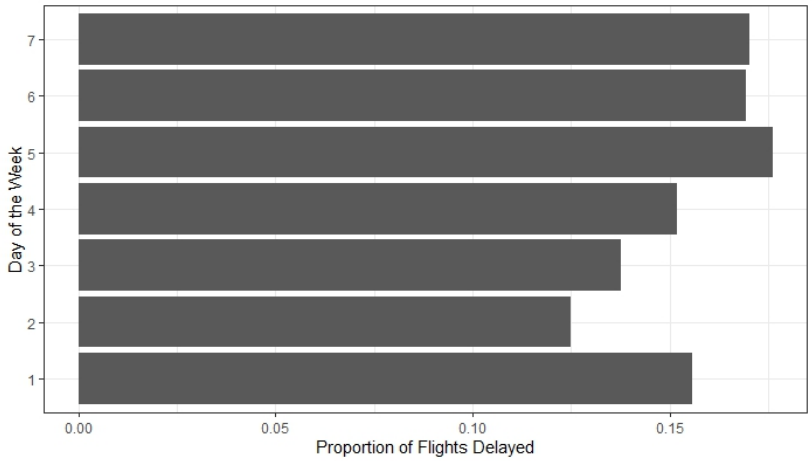

Figure 1: The proportion of flights that were delayed on each day of the week, with 1 being Monday and 7 being Sunday.

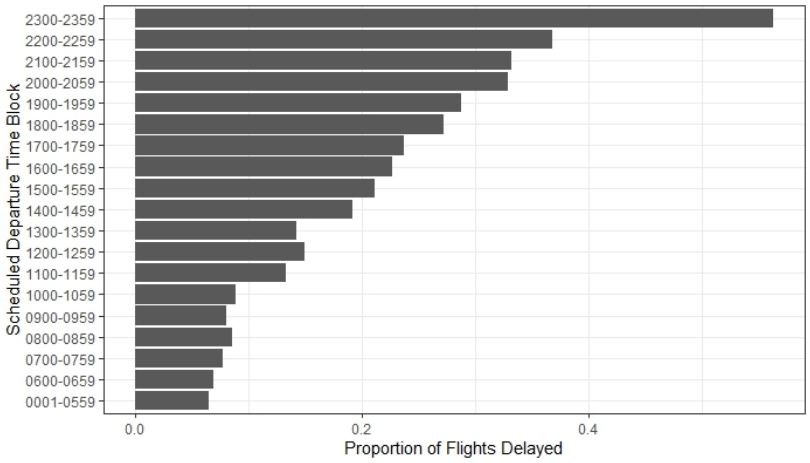

Figure 2: The proportion of flights that were delayed for each time block.

We also believed that different airlines and locations have differing delay rates. Figure 3 shows the delay rates for each distinct carrier code and reveal that Spirit (NK) and Frontier (F9) flights are delayed the most often, which align with our experiences. Figure 4 shows that flights going to states in the western part of the country, like California, Nevada, and Utah, were delayed more frequently. Our observations in this section formed the basis of our feature engineering.

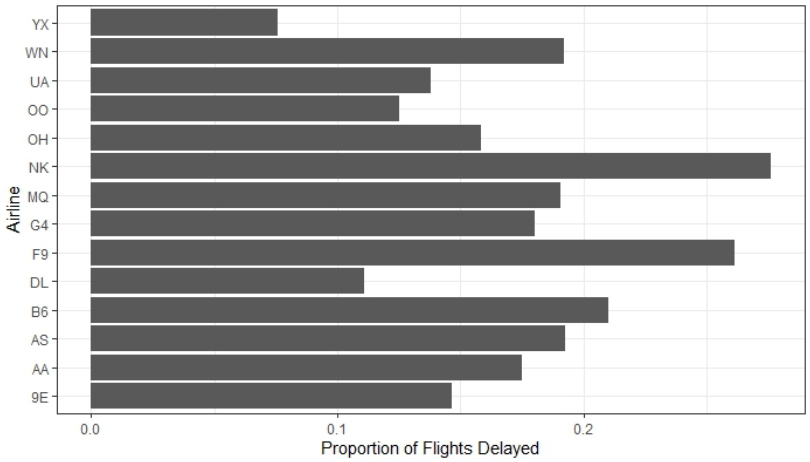

Figure 3: The proportion of flights that were delayed for each airline.

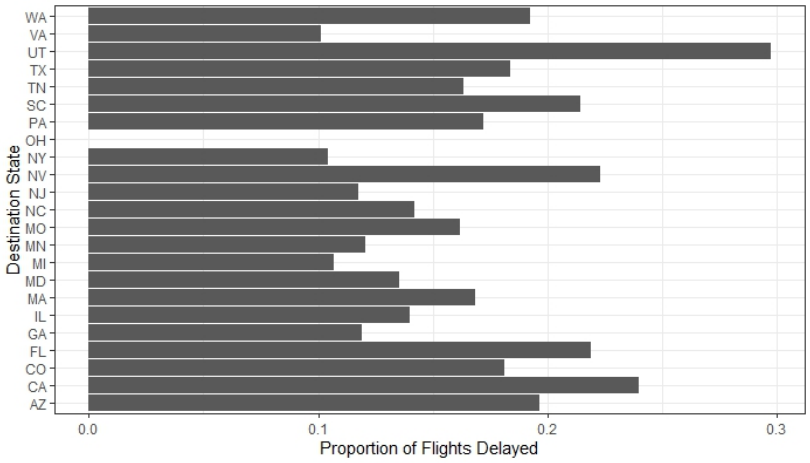

Figure 4: The proportion of flights that were delayed for each destination state.

3 Supervised Analysis

The prediction follows a binary classification problem, where the response variable is DEP_DEL15, indicating whether there was a delay of greater than 15 minutes or not. Rather than directly predicting class labels, our models output predicted probabilities of delay, which are then evaluated using the AUC metric.

3.1 Training, Validation & Testing

Because the prediction task is time dependent, we adopted a validation framework that reflected this, rather than implementing a random train-validation split. Data from 2023–2024 was used for training, while the 2025 "visible" dataset served as a validation set. This setup mimics a realistic prediction setting, where only past data is available when making predictions on future flights. The final test data was the 2025 "guess" dataset, and the 2025 "visible" flights were used to calculate same day features for this.

3.2 Feature Engineering

A key concern throughout this process was avoiding data leakage. First, we removed all cancelled and diverted flights from the data, since these are highly irregular events that could impact the calculation of our features. All features were carefully constructed to ensure that only information available prior to a flight's scheduled departure time was used. In particular, rolling or cumulative features (such as same day delay rates) were computed using only earlier flights within that day. Our approach included features that can be grouped into four categories:

3.2.1 Historical Predictors

Rather than using several categorical variables with a large number of categories (such as carrier) directly, which led to computational inefficiency in earlier attempts, we constructed aggregated delay rates from the training data. These include:

1. Carrier-level delay rate: the overall proportion of flights operated by a given airline that are delayed, reflecting its general reliability.
2. Carrier–route delay rate: the proportion of flights delayed for a specific airline on a specific route (origin–destination pair), capturing route-specific performance.
3. Destination-level delay rate: the proportion of flights delayed when flying to a given destination, indicating congestion at the arrival airport.

4. Carrier–month delay rate: the proportion of flights delayed for a given airline within a specific month, capturing seasonal patterns.

This transformation reduces dimensionality while preserving meaningful structure, allowing models to capture long-term reliability patterns without incurring large computational costs.

3.2.2 Time Dependent Predictors

Our exploratory analysis suggested that delays vary systematically with time of day, week and year, so we constructed:

1. Hours since midnight: The scheduled departure time is converted into a continuous variable representing hours since midnight.
2. Long Weekends: Flights are grouped into "long weekends" (Friday through Monday) versus weekdays, since the EDA suggested that delays tend to follow similar patterns Friday through Monday.
3. Season (Winter, Spring, Summer, Fall): Months are grouped into seasons to account for seasonal patterns in delays, such as increased weather related disruptions in winter or higher traffic during summer. The breakdown of this variable is as follows:
   (a) Winter: December, January, February
   (b) Spring: March, April, May
   (c) Summer: June, July, August
   (d) Fall: September, October, November

3.2.3 Same-Day Predictors

To capture real time conditions, we engineered features based on earlier flights on the same day:

1. Carrier-day delay rate: proportion flights for a carrier that were delayed earlier that day.
2. Previous 3 hour weather delay: rolling proportion of weather-related delays in the three hours prior to scheduled departure.

These features are particularly important for capturing the more transient effects of carrier and weather which might not be reflected in historical averages.

3.2.4 Aircraft Level Predictors

These features aim to understand the delays that take place due to an aircraft's schedule. To capture this, we used arrival data to construct:

1. Is First: indicator for whether the flight is the aircraft's first departure of the day.
2. Is Incoming Delayed: whether the aircraft's previous flight arrived late.
3. Turnaround Time: time between the previous arrival and the current departure.

These features are motivated by operational constraints: if an aircraft arrives late and has a short turnaround time, the subsequent departure is more likely to be delayed. Interaction terms between the Is First variable and the other aircraft level features were included to account for the fact that turnaround time and incoming delay are not meaningful for first flights of the day.

3.3 Final Dataset

The final features selected for modeling are presented in Table 1.

Feature	Description	Type
CARRIER_DAY_DELAY_RATE	Proportion of earlier flights on the same day for a given carrier that were delayed, capturing real-time performance.	Numerical
CARRIER_DELAY_RATE	Historical proportion of delayed flights for a given airline, reflecting overall carrier reliability.	Numerical
CARRIER_MONTH_DELAY_RATE	Historical delay rate for a carrier within a given month, capturing seasonal airline effects.	Numerical
CARRIER_ROUTE_DELAY_RATE	Historical delay rate for a specific airline on a specific route, capturing route-level reliability.	Numerical
DEP_DEL15	**Target variable:** indicator for whether the departure delay is at least 15 minutes.	Categorical
DEP_HOURS_SINCE_MIDNIGHT	Scheduled departure time expressed as continuous hours since midnight to capture time-of-day effects.	Numerical
DEST_DELAY_RATE	Historical proportion of delays for flights to a given destination, reflecting destination congestion.	Numerical
DEST_STATE_ABR	State abbreviation of the destination airport, capturing geographic variation in delays.	Categorical
DISTANCE	Flight distance in miles, representing route length and potential exposure to delays.	Numerical
IS_FIRST	Indicator for whether the flight is the aircraft's first departure of the day.	Categorical
IS_INCOMING_DELAYED	Indicator for whether the aircraft's previous flight arrived delayed, capturing delay propagation.	Categorical
SEASON	Categorical grouping of month into Winter, Spring, Summer, or Fall to capture seasonal patterns.	Categorical
TURNAROUND_TIME	Time between the aircraft's previous arrival and current departure, reflecting schedule tightness.	Numerical
WEATHER_DELAY_RATE_3HR	Rolling proportion of weather-related delays in the previous three hours.	Numerical
YEAR	Calendar year of the flight, included to capture temporal differences across years.	Numerical

Table 1: Final set of features used in the model, with **DEP_DEL15** as the target variable.

The relationship between the predictors and the target variable reflect a mix of long-term, same-day and aircraft-specific effects. Time dependent variables like year, season and hours since midnight capture more broad trends, with delays varying across years, being more common in certain seasons, and generally increasing later in the day as delays build up. Route related features such as the destination state and distance help account for the fact that some routes or regions are more prone to delays than others. Historical delay rates, specifically carrier delay rate, carrier route delay rate, destination delay rate and carrier month delay rate, also play an important role, with higher past delay rates corresponding to higher chances of delay in the present. In addition to this, same day features like carrier day delay rate and weather delay rate in the last 3 hours help capture what's happening in real time, such as congestion or weather disruptions earlier in the day. Aircraft level features are also incredibly important, as flights that are not the first of the day and have a delayed incoming aircraft are much more likely to be delayed, particularly when the turnaround time is short, since there's less time to recover from earlier delays. We believe that each of these features can have a strong impact in predicting the final delay.

3.4 Models and Tuning

We evaluated several classification models with increasing flexibility:

3.4.1 Logistic Regression

A baseline logistic regression model was fit with all features and selected interaction terms. This model achieved an AUC of approximately 0.733. Even though it's a simple model, it performed quite well, suggesting that the features we engineered already capture important patterns.

3.4.2 Regularized Logistic Regression (Ridge and LASSO)

To address potential multicollinearity and improve generalization, we fit both a Lasso (L1 regularization) and Ridge (L2 regularization) model. The regularization parameter was selected via cross-validation. Both models achieved similar performance (AUC $\approx$ 0.732–0.733), indicating that there is limited benefit from feature selection and there is no severe multicollinearity among our selected predictors.

3.4.3 Random Forest

A random forest model was trained using the ranger implementation, with hyperparameters (such as mtry, min.node.size) tuned via cross-validation. This model achieved an improved AUC of approximately 0.748. The improvement suggests that there in fact presence of nonlinear relationships and interactions among features that are not fully captured by previously tried linear models.

3.4.4 XGBoost

Finally, we trained a gradient boosting model using XGBoost, with hyperparameters once again selected via cross-validation and early stopping. This model achieved the best performance, with an AUC of approximately 0.759. Boosting improves performance by sequentially focusing on previously misclassified observations, allowing it to capture more subtle patterns and interactions in the data.

Overall, linear models performed well due to strong feature engineering, while Random Forest improved performance by capturing nonlinearities. Boosting provided the best results by better refining these nonlinear patterns.

3.4.5 Final Model

The final model we selected was **XGBoost**, based on its AUC being higher than other models.

The XGBoost results show that aircraft level and real time features matter the most. In particular, turnaround time and whether the incoming flight was delayed are the strongest predictors, highlighting how delays carry over between flights. Time of day (represented by hours since midnight) also plays a big role, with delays building up later in the day. Same day features like carrier day delay rate and recent weather delays also contribute, while broader features like destination state or long term averages matter less. Overall, this suggests that immediate conditions are more important than general historical patterns in predicting delays.

4 Analysis of Results

4.1 Which Flights Did We Do Well On?

Our final model overpredicted delays: about 23% of flights in the test data were actually delayed, but our average prediction probability was 0.28. Figure 5, which compares our XGBoost probability buckets with observed delay frequency reveals a much bigger problem: our model might as well have been assigning every flight the same probability of delay and is not well-calibrated at all, a result confirmed by an AUC of 0.5047 on the true labels.

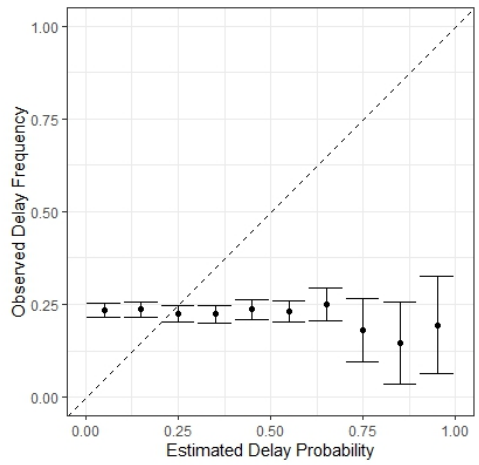

Figure 5: A calibration plot which compares our fitted delay probabilities to actual delay frequency.

4.2 Predicting Minutes Delayed

If the goal were to predict delay minutes instead of a binary outcome, we would adjust our approach to account for the fact that the data is highly skewed, with most flights having no delay. We suggest a two-stage approach:

- Stage 1: Use a classification model to predict whether a delay occurs (where a delay is defined as greater than 15 minutes).
- Stage 2: For flights predicted to be delayed, use a regression model to estimate the number of minutes delayed.

In this setup, the second model is trained only on flights that actually experienced delays, which helps prevent it from being dominated by the majority class (non-delayed flights). For the regression, we could use Poisson regression, since delay minutes are non-negative values which Poisson handles adeptly. Although there are some negative values in the original delay variable to indicate flights taking off earlier than scheduled, this is not an issue here because we restrict the training data to observations with delays greater than 15 minutes.

4.3 Tradeoff

If, for our own planning purposes, we cared about a tradeoff in cost between missing a flight C and showing up on-time for a delayed flight r, we could adapt our strategy by implementing an asymmetric loss function, which would shift the classification decision boundary (which we did not explicitly have, since we used probabilities) away from 0.5. A lower C/r means that we care more about waiting for a long time than missing the flight, so we would increase the decision boundary and show up to the airport later. On the other hand, a higher C/r means that missing the flight would be very costly, so we would decrease the decision boundary and show up to the airport earlier or on-time.

4.4 Further Analysis: Limitations

It must be noted that the relatively small gap between logistic regression and more complex models suggests that feature quality might be more important than model complexity in this problem. Additionally, the dataset is imbalanced, with only 15.52% of flights experiencing delays. This means that even a simple model that predicts no delay for every flight would already achieve fairly high accuracy, so classification rate is not very informative here. In practice, our models only improve on this baseline by a small margin, suggesting that while the engineered features do capture some signal, there is still room for improvement. Going forward, better feature engineering is likely to be more helpful than increasing model complexity.